

SHAKTHI S

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EDUCATION

Bachelor of Technology, Electrical and Electronics Engineering
Amrita Vishwa Vidyapeetham | CGPA: 8.07

Oct 2022 – Present
Coimbatore, India

Bachelor of Science, Data Science and Applications
Indian Institute of Technology, Madras

Jan 2023 – Present
Online

SKILLS

Languages: Python, C, HTML5, CSS

Libraries & Frameworks: Pandas, NumPy, TensorFlow, TensorFlow Lite, PyTorch, Keras, scikit-learn, Flask, SQLAlchemy

Databases & Cloud: MySQL, Firebase

Tools: Git, GitHub, VS Code, MATLAB

Embedded & Hardware: ESP32, Arduino, Raspberry Pi

Operating Systems: Kali Linux, Raspberry Pi OS, Linux

Domains: IoT Systems, Edge AI, REST API Integration, Network Protocols (MQTT)

PROFESSIONAL EXPERIENCE

Internship Trainee, Satish Dhawan Space Centre SHAR, ISRO

May 2025 – Jun 2025
Sriharikota, India

- Analysed Multi-Object Tracking Radar (MOTR) and antenna arrays supporting real-time signal transmission for rocket launches.
- Authored technical documentation, system workflow diagrams, and safety protocols for mission-critical launch operations.

In-plant Trainee, Kudankulam Nuclear Power Project, NPCIL

Dec 2024
Kudankulam, India

- Performed diagnostics on high-voltage systems (transformers, circuit breakers); prepared maintenance & compliance reports under nuclear safety standards.

PROJECTS

Detection of False Data Injection Attacks in Smart Micro-Grid Using Edge AI

Implemented a ML-based cyber attack detection using ESP32–Raspberry Pi MQTT communication, with Firebase-based cloud logging and real-time monitoring dashboards for a three bus system.

Predictive Analysis of YouTube Videos Using Neural Networks

Analysed YouTube video engagement metrics via YouTube Data API and VADER sentiment analysis; built a neural network model achieving 86.72% accuracy in predicting ad-relevant performance indicators — surfacing actionable insights for content optimization.

VisionSole - Smart Assistant for the Visually Impaired

Developed a wearable assistive system using ESP32-CAM and sensors for real-time obstacle detection, integrated with an Android app using Google Maps API for voice navigation and emergency alerts.

Smart Vehicle Parking Web Application

Developed and deployed a Flask-based parking management system with role-based access, real-time slot booking, and billing workflows using SQLAlchemy ORM with SQLite.

IoT-Based Plant Leaf Health Monitoring

Trained & deployed CNN (73.5% accuracy) for leaf disease detection on ESP32 via TFLite; mobile dashboard alerts cut manual inspection ~50%.

PUBLICATIONS

Breast Cancer Screening from Thermal Images: CNN-Based Analysis with Augmented Datasets and Web App Deployment - IEEE INSPECT 2025

Predictive Modeling of Video Streaming Effectiveness Using Artificial Neural Network and Sentiment Analysis - IEEE ICERECT 2025

Spectrum-Agnostic Space Debris Detection with Autoencoders - IEEE ICAUC 2026

CERTIFICATIONS

Foundation in Data Science and Applications - **IIT Madras** | Artificial Intelligence and Edge Computing - **L&T Edutech** | MATLAB OnRamp - **MathWorks**

EXTRA-CURRICULARS

Volunteer - **National Service Scheme (NSS)** | Electronics Engineer - **Amrita Robotics Club** | Social Media Team Designer - **Amrita Animal Welfare Society**